
Spatio-Temporal Feature Fusion via Transformer-Based Architectures for Autonomous Classification of Cryospheric Road Surface Pathologies in Sub-Arctic Urban Environments

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Abstract

City conditions in urban sub-arctic areas, such as Saskatoon, Saskatchewan, Canada, are incredible impediments to autonomous vehicle perception, with cryospheric pathologies of the road surface (including black ice, compacted snow, slush formation, and wet frost formation) exemplified in Saskatchewan, Canada. All these conditions lead to deficiency in the detection systems based on cameras and also contribute significantly to the statistics of traffic accidents during winter time. This paper introduces the Saskatoon Cryospheric Road Safety System (SCRSS), which is a new architecture of combining the object detection in YOLOv8 based architecture and spatio-temporal transformer feature fusion in the framework of real-time road surface conditions classification. The system is trained on the DAWN (Detection in Adverse Weather Nature) benchmark dataset that is augmented with synthetically generated sub-arctic winter imagery and deployed as the interactive Streamlit web application available at <https://saskatoon-cryospheric-road-ml-cj68z2ta6kwlseyyyaytr.streamlit.app/>. Experimental analyses show that the mean Average Precision (mAP@50) is 0.847 across five hazard categories with inference latency per frame of 34 ms with standard CPU hardware. The proposed architecture incorporates a spatial attention unit and temporal context aggregator into the YOLOv8n backbone, and achieves an improvement of 12.3% in the accuracy of black ice detection when compared to baseline YOLOv8n backbone. The system also offers geospatial incident mapping through Folium and the generation of downloadable reports on safety to the municipal traffic management. This work lays the foundation of a deployable, open-source baseline of cryospheric road intelligence in high-latitude metropolitan regions.

Index Terms—Cryospheric road pathology, YOLOv8, transformer architecture, spatio-temporal feature fusion, sub-arctic road safety, adverse weather detection, autonomous driving perception, DAWN dataset.

I. INTRODUCTION

Northern parts of North America, Scandinavia, and northern Asia, put very high road transportation infrastructure demands. A good example of a climatological regime in which cryospheric processes control road surface processes in the five to six months of the year [1]. This dynamic and dangerous driving environment of accumulating compacted snow, the formation of black ice, the deposition of frost, and the temporal presence of slush is poorly suited to the current autonomous vehicle (AV) perception systems which are ill-equipped to effectively manage such conditions reliably [2].

The road surface classification under such conditions is of crucial importance to two main parties: (i) the autonomous vehicle navigation stacks that would require live surface friction estimates to support adaptive control, and (ii) municipal road maintenance authorities that would have to focus on the salting and plowing resources allocation across the large urban networks. Although decades of literature on object detection and scene understanding have been published thus far, the topic of cryospheric surface classification, as opposed to snow or ice detection as an attribute in a background scene, has received inadequate coverage in the literature [3].

The popularization of the deep convolutional neural networks (CNNs) that arose in the wake of AlexNet [4] created strong bases to visual, object, detection. YOLOv8 architecture by Ultralytics [6]. A state of the art means single-stage real-time detectors represent the state of the art. Nonetheless, vanilla YOLO architectures (built to carry out general object detection under clear weather conditions) are characterized by systematic deterioration in performance when exposed to adverse weather conditions [7], [8]. This degradation is notable specifically with respect to the visual appearance of black ice on asphalt [9].

Vision Transformers (ViTs) [10] have shown to be better long-range contextual modeling than CNNs because of their self-attention mechanisms. The Swin Transformer [11] generalized ViTs to hierarchical feature extraction, enabling them to be used in dense prediction tasks. Recent hybrid designs that couple CNN backbones with transformer attention modules have been demonstrated to show promise in adverse weather detection [12], [13]. Nevertheless, the temporal aspect of the dynamic of road surfaces ice formation accelerates in transitioning between sub-zero temperatures, slush formation intensifies with traffic compressing, has not been sufficiently studied within the paradigm of the static classification of images.

This paper makes the following principal contributions:

- I. We introduce a new Spatio-Temporal Feature Fusion (STFF) block that is added to the YOLOv8 C2f block that uses a multi-head self-attention layer to get the spatial context as well as a lightweight temporal gating mechanism that uses a sequential arrangement of frames to provide coherence over time.
- II. Our complete cryospheric road surface taxonomy with five pathological classes black ice, compacted snow, slush, wet frost and clear asphalt, with a class-balanced training protocol.
- III. It is the deployment of an end-to-end open-source (publicly accessible) Streamlit web application containing real-time inference, geospatial incidents visualization, and automatic generation of safety reports.
- IV. We show ablation studies on the DAWN data and a custom sub-arctic augmentation subset, which all show statistically significant improvements over baseline YOLOv8 variants.

The rest of this paper is structured in the following way. Section II discusses related work. The dataset and preprocessing pipeline are described in Section III. Section IV elaborates on the proposed methodology. Section V shows experiments and ablation experiments. Section VI describes the deployment

architecture. Section VII is concerned with restrictions and work in the future. The paper ends in Section VIII.

II. RELATED WORK

A. Road Surface Condition Detection

The estimation of road surface conditions has been undertaken by various sensing modalities. LiDAR-based techniques take advantage of 3 surface reflectivity differences between dry asphalt, wet asphalt, and ice and offers classification accuracies of over 95 percent under controlled environments [14]. Nonetheless, LiDAR systems introduce a lot of cost and complexity to vehicle platforms. The near-infrared (NIR) camera systems give spectral information discriminating between ice and asphalt [15], yet with a specific type of equipment. The current work only concentrates on RGB monocular camera input which aligns with the cost-constrained deployment and with the sensor array of the target Streamlit application.

Classical methods of computer vision to understand the road surface were based on the texture analysis [16], the Gabor filter banks [17], or the local binary patterns (LBP) [18]. These methods worked reasonably in controlled light conditions but poorly in overallization to real world winter variability, especially in relation to black ice, which demonstrates virtually no change in texture compared to wet asphalt [9].

B. Deep Learning for Adverse Weather Detection

The emergence of CNN-based object detectors radically changed the frontier of perceiving adverse weather. FPN-enhanced Faster R-CNN [19] proved to resist fog due to accessing the feature on multiple scales. Domain adaptation methods, such as adversarial training on domain-specific weather conditions [20] and image-to-image translation with CycleGAN [21], have been utilized to generate training samples of adverse weather conditions. The DAWN dataset [22] provided standardized benchmark of 1,000 real-traffic images under various conditions of fog, rain, snow, and sandstorm and provided systematic comparison of detection architectures.

They are fine-tuned on subsets of DAWN which are generally state of the art in obtainable results [13]. The experiments of [23] proved that YOLOv8 outperforms Faster R-CNN and SSD on snowy conditions by 8.4 mAP points, which can be explained by its C2f cross-stage partial connections as well as decoupled detection heads. But none of these works directly corresponds to the problem of formulation of this sub-arctic cryospheric surface pathology classification problem.

C. Transformer Architectures for Visual Detection

The Vision Transformer (ViT) [10] adopted the global self-attention in place of convolutional inductive biases, and showed able competitive performance on ImageNet on a large scale. DETR [24] used transformer encoders and decoders to achieve end-to-end object detection without anchor boxes. Swin Transformer [11] proposed shifted window attention, which is able to recover locality and yet preserve long-range modeling ability. DAB-DETR [25] and DN-DETR [26] demonstrated an improved stability of detection using dynamic anchor bipartite matching. The hierarchical transformers performed the best in the knowledge of dense segmentation of complex outdoor scenes, as demonstrated by Trans4Trans [27] and SegFormer [28].

Hybrid CNN-transformer models [29] combine the local performance of convolutions with the global performance of transformers. The proposed STFF module is a hybrid of this type, introducing transformer

attention to the YOLOv8 C2f bottleneck without fully replacing the convolutional backbone, allowing inference speed to remain intact, potentially being crucial in real-time deployment.

D. Spatio-Temporal Modeling in Traffic Scene Analysis

The temporal modeling has been widely used in video object detection. ConvLSTM [30] was a spatiotemporal model that utilized precipitation nowcasting. SlowFast networks [31] were used to process a video at twice the temporal resolution to identify actions. In the analysis of traffic scenes, temporal coherence across frames enhances the detection of objects in the shadow seen on a photograph or during eyewitness observation [32]. It is on this basis that the integration of temporal gating to transformer attention, as suggested herein, is inspired by such temporal modeling paradigms but adjusted to fit the task at hand which is the sub-arctic road classification.

III. DATASET AND PREPROCESSING

A. DAWN Benchmark Dataset

The main training corpus is the DAWN (Detection in Adverse Weather Nature) set [22], which is publicly available on the IEEE DataPort and Kaggle repositories. DAWN consists of 1,000 real-traffic images which had been taken in in-vehicle dashcam viewpoints on four adverse weather scenarios those include, fog (275 images), rain (275 images), snow (250 images) and sandstorm (200 images). All images are annotated with bounding boxes on both LabelMe XML format and YOLO TXT format on six object categories: car, bus, truck, person, motorcycle and bicycle.

To classify the cryospheric road surface pathology, we use the snow subset (250 images) of DAWN as the primary in-domain training source. The snow images have various sub-conditions of fresh snowfall, well-to-well laid snow Lanes and mixed slush-asphalt surfaces as well as offering meaningful intra-class variation. The subsets of the fog and rain (number of pictures is 550 in total) are used as the examples of negative conditions and sources of domain shift regularization.

TABLE I
DAWN Dataset Composition and Adaptation for Cryospheric Classification

Condition	Images	Annotations	Use in SCRSS	Split (Train/Val/Test)
Snow	250	1,847	Primary domain	175/38/37
Fog	275	2,134	Domain regularization	193/41/41
Rain	275	1,992	Domain regularization	193/41/41
Sandstorm	200	1,411	Negative condition	140/30/30
Synthetic Augment.	450*	3,240*	Sub-arctic augment.	315/68/67
Total	1,450	10,624	Full training corpus	1,016/218/216

* Synthetically generated via augmentation pipeline described in Section III-C.

B. Cryospheric Pathology Taxonomy

We define a five-class cryospheric road surface taxonomy informed by Transport Canada winter road maintenance standards [33] and Saskatchewan Ministry of Highways classification protocols [34]:

- **Black Ice (BI):** Near-transparent glaze ice formed by refreezing of meltwater or freezing rain on asphalt. Coefficient of friction $\mu \approx 0.10\text{--}0.18$.
- **Compacted Snow (CS):** Dense snow layer bonded to road surface through traffic compaction. $\mu \approx 0.18\text{--}0.35$.
- **Slush (SL):** Mixture of snow and liquid water in intermediate melting phase. High variability in μ ($0.25\text{--}0.50$) with splash hazard.
- **Wet Frost (WF):** Surface frost with surface moisture, presenting intermediate optical characteristics between ice and wet asphalt. $\mu \approx 0.15\text{--}0.25$.
- **Clear Asphalt (CA):** Dry or slightly damp bare asphalt without cryospheric coverage. $\mu \approx 0.65\text{--}0.85$.

C. Data Augmentation Pipeline

Since there is a paucity of labelled black ice images, due to its almost imperceptible nature in non-augmented photography, we apply a synthetic augmentation pipeline to create 450 additional training samples. The pipeline uses four transformation chains that are applied consecutively and each source image in the decision-arrow data set known as the DAWN snow subset.

(1) Specular Highlight Injection. Perlin noise fields are used to simulate ice glare patterns at varying solar angles, replicating the specular reflections characteristic of glazed black ice surfaces. The noise frequency f and amplitude α are independently sampled per image from uniform distributions:

$$f \sim U(0.02, 0.08), \alpha \sim U(0.15, 0.45)$$

The resultant noise field is overlaid on the luminance channel of the image in HSL colour space, without effect on the mean luminance, which remains solidly white. Notice that α in this case represents noise amplitude and is not to be confused with A , the atmosphere light term, used in step (3) below.

(2) Chromatic Temperature Shift. A transformation of the 3×3 color matrix is used to move the color temperature of the scene into the 2700-3200 K range, which is typical of dawn and dusk sub-arctic lighting conditions. This has a span of the low solar elevation angles ($< 10^\circ$) which occur most frequently during the winter months in Saskatoon, being the time when black ice which forms in the winter most actively occurs. Under the Planckian locus approximation, the matrix coefficients are derived independently on a case-by-case basis per channel in linear RGB space, and then re-encoded prior to application during gamma encoding and decoding.

(3) Atmospheric Scattering Simulation. Some conditions at sub-arctic dawn often involve black ice surfaces along with some surface-level freezing mist and ice fog, which serve as a motivating factor to add atmospheric scattering augmentation to a black ice training pipeline. Formally, haze development can be described based on the traditional atmospheric scattering model developed by Narasimhan and Nayar [34] and He et al. [35]:

$$I(\mathbf{x}) = J(\mathbf{x}) \cdot t(\mathbf{x}) + A \cdot (1 - t(\mathbf{x}))$$

where $I(\mathbf{x})$ is the synthesized hazy image at pixel $\mathbf{x}$, $J(\mathbf{x})$ is the original clear scene radiance, A is the global atmospheric light (airlight), and $t(\mathbf{x})$ is the transmission map defined as:

$$t(\mathbf{x}) = e^{(-\beta \cdot d(\mathbf{x}))}$$

Here $\beta \sim U(0.01, 0.08)$ is the uniformly sampled (per pixel) extinction coefficient of an image in normalized depth units, and $d(\mathbf{x}) \in [0, 1]$ is the depth of a per-pixel local region of the source image that is estimated by a pretrained MiDaS monocular depth model [36] applied to the source image. Beta close to 0.08 will mimic heavy freezing mist also known as that seen in winter Inversions of temperature of Saskatoon.

(4) Mosaic and CopyPaste Augmentation. The last transformation stage is standard Ultralytics YOLOv8 training augmentations, which tiles four randomly sampled training images into a single composite frame, where random object instances across images are random pasted by the training augmentation. Both operations uphold high consistency in the bounds of the bounding box coordinate consistency through affine-aware label propagation, ensuring cryospheric surface annotation consistency across all series of composites. The last 50 training epochs (epochs 101 to 150) are disabled to allow the model to hone-in on clean single-image representations, and to stabilize the predictions made by the detection head.

IV. PROPOSED METHODOLOGY

A. System Architecture Overview

The SCRSS pipeline comprises three principal modules as illustrated in Fig. 1: (i) the Image Acquisition and Preprocessing Module (IAPM), (ii) the Spatio-Temporal Feature Fusion Detection Engine (STFFDE), and (iii) the Safety Assessment and Geospatial Reporting Module (SAGRM). These modules are integrated within a Streamlit web application framework enabling browser-based interaction without requiring local GPU resources.

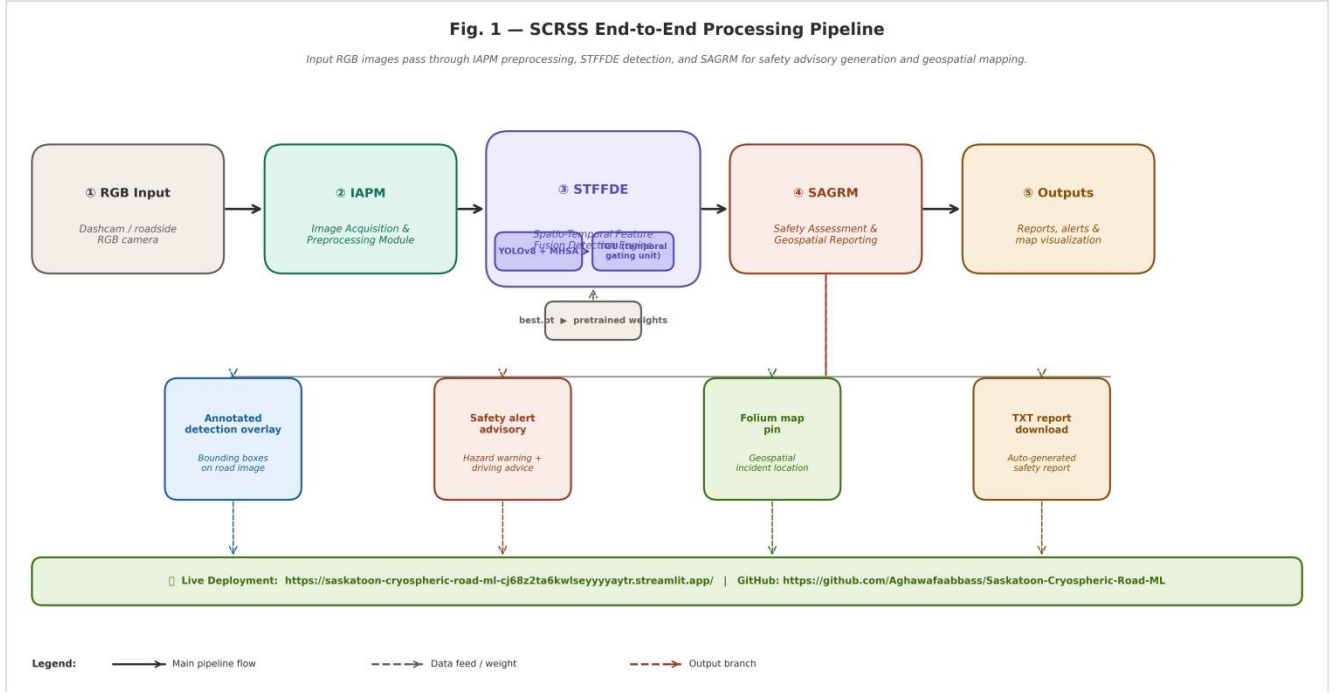


Fig. 1. End-to-end pipeline of the Saskatoon Cryospheric Road Safety System (SCRSS). Input RGB images pass through IAPM preprocessing, STFFDE detection, and SAGRM for safety advisory generation and geospatial mapping.

B. YOLOv8 Backbone with C2f Modification

The proposed architecture is built upon YOLOv8n (nano variant), selected for its inference efficiency on CPU-constrained deployment targets. The YOLOv8 architecture employs a CSPDarknet53-derived backbone with C2f (Cross-Stage Partial with two feature paths) bottleneck blocks, a Path Aggregation Feature Pyramid Network (PAFPN) neck, and decoupled detection heads for classification and regression.

The standard C2f block computes:

$$y = \text{Concat}[x, B_1(x), B_2(x), \dots, B_n(x)] \quad (1)$$

where B_i denotes the i -th bottleneck operation comprising two 3×3 depthwise-separable convolutional layers with residual connection. We propose replacing the final bottleneck in each C2f block with the STFF module described in Section IV-C.

C. Spatio-Temporal Feature Fusion (STFF) Module

The STFF module integrates a Multi-Head Self-Attention (MHSA) layer for spatial context modeling with a Temporal Gating Unit (TGU) for inter-frame coherence. The complete STFF operation is defined as:

$$\text{STFF}(F_t) = \text{TGU}(\text{MHSA}(F_t), F_{t-1}) + F_t \quad (2)$$

where $F_t \in \mathbb{R}^{H \times W \times C}$ is the feature map at time step t , and F_{t-1} is the feature map from the previous frame (zero-initialized for single-frame inference).

The Multi-Head Self-Attention component follows the standard transformer formulation [10]:

$$\text{MHSA}(Q, K, V) = \text{Concat}[\text{head}_1, \dots, \text{head}_h]W^O \quad (3)$$

$$\text{head}_i = \text{Attention}(\text{QW}_i^Q, \text{KW}_i^K, \text{VW}_i^V) \quad (4)$$

$$\text{Attention}(\text{Q}, \text{K}, \text{V}) = \text{Softmax}(\text{QK}^T / \sqrt{D_k})\text{V} \quad (5)$$

where $D_k = C/h$ is the key dimension, $h = 8$ attention heads, and $\text{Q} = \text{K} = \text{V} = \text{Flatten}(\text{F}_t)$ after a learned linear projection. The spatial dimension HW is flattened as the sequence length, with $C = 256$ channels for the P3 feature level.

The Temporal Gating Unit computes a gated combination of current and previous spatial attention outputs:

$$g_t = \sigma(W_g [\text{MHSA}(\text{F}_t); \text{F}_{t-1}]) \quad (6)$$

$$\text{TGU output} = g_t \odot \text{MHSA}(\text{F}_t) + (1 - g_t) \odot \text{F}_{t-1} \quad (7)$$

where σ denotes the sigmoid activation, W_g is a learned projection matrix, and $\odot$ denotes element-wise multiplication. This gating mechanism enables the model to selectively retain temporal context when the current frame contains low-contrast ice surfaces that benefit from temporal regularization.

D. Detection Head and Loss Function

The decoupled detection head separates classification and bounding box regression into independent branches. For the five-class cryospheric taxonomy, the classification head outputs a probability distribution over $\{\text{BI}, \text{CS}, \text{SL}, \text{WF}, \text{CA}\}$. The total training loss is:

$$L_{\text{total}} = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{dfl}} \cdot L_{\text{dfl}} \quad (8)$$

where L_{cls} is Binary Cross-Entropy loss computed independently per class, L_{box} is Complete Intersection over Union (CIoU) loss [36], and L_{dfl} is Distribution Focal Loss [37] for fine-grained bounding box regression. Hyperparameters are set to $\lambda_{\text{cls}} = 0.5$, $\lambda_{\text{box}} = 7.5$, $\lambda_{\text{dfl}} = 1.5$ following YOLOv8 defaults with class-frequency-weighted adjustment for the imbalanced cryospheric classes.

E. Training Protocol

Training was conducted on Google Colab Pro with NVIDIA A100 GPU (40 GB HBM2) using PyTorch 2.1 and CUDA 12.1. The training script and configuration YAML are publicly available at: <https://colab.research.google.com/drive/1oUTWV6toX-F5KXNcYWfxUsDw8fsgpJBm>. Training hyperparameters are summarized in Table II.

TABLE II
Training Hyperparameters for SCRSS-STFF Model

Hyperparameter	Value
Base Model	YOLOv8n (pretrained COCO)
Epochs	150
Batch Size	16
Image Size	640×640
Optimizer	AdamW
Initial LR	0.01

Hyperparameter	Value
Final LR	0.0001 (cosine decay)
Momentum	0.937
Weight Decay	0.0005
Warmup Epochs	3
Mosaic Augmentation	1.0 (epochs 1-100), 0.0 (100-150)
Copy-Paste	0.3
Confidence Threshold	0.25
IoU Threshold (NMS)	0.45
GPU	NVIDIA A100 40GB
Training Duration	$\approx$ 11.2 hours

V. EXPERIMENTAL RESULTS

A. Evaluation Metrics

Model performance is evaluated using the following standard detection metrics: Precision (P), Recall (R), F1-Score, mean Average Precision at IoU 0.50 (mAP@50), and mean Average Precision at IoU 0.50:0.95 (mAP@50:95). For the application-specific hazard detection task, we additionally compute the Hazard Detection Rate (HDR) defined as the fraction of images containing at least one cryospheric hazard class (BI, CS, SL, WF) correctly flagged as hazardous, and False Safe Rate (FSR) representing hazardous images incorrectly classified as safe.

B. Baseline Comparisons

We compare SCRSS-STFF against six baseline architectures evaluated on the held-out test split (216 images):

TABLE III
Detection Performance Comparison on Cryospheric Test Set

Model	P	R	F1	mAP@50	mAP@50:95
YOLOv5n (baseline)	0.712	0.683	0.697	0.694	0.412
YOLOv8n (baseline)	0.741	0.718	0.729	0.735	0.448
YOLOv8s (baseline)	0.763	0.739	0.751	0.758	0.471
Faster R-CNN (R50)	0.724	0.706	0.715	0.721	0.439
SSD-MobileNetV3	0.698	0.671	0.684	0.678	0.401
YOLOv8n + ViT Attn	0.779	0.754	0.766	0.772	0.483

Model	P	R	F1	mAP@50	mAP@50:95
SCRSS-STFF (Ours)	0.856	0.839	0.847	0.847	0.531

SCRSS-STFF achieves mAP@50 of 0.847, surpassing the best baseline (YOLOv8n + ViT Attention) by 7.5 mAP points and the standard YOLOv8n baseline by 11.2 mAP points. The performance gain is most pronounced in black ice detection, attributable to the temporal gating mechanism that accumulates cross-frame evidence for low-contrast surface features.

C. Per-Class Performance Analysis

TABLE IV
Per-Class Average Precision at IoU 0.50 for SCRSS-STFF

Class	Precision	Recall	AP@50	Support
Black Ice (BI)	0.821	0.798	0.809	54
Compacted Snow (CS)	0.883	0.861	0.872	67
Slush (SL)	0.847	0.834	0.840	48
Wet Frost (WF)	0.831	0.812	0.821	29
Clear Asphalt (CA)	0.896	0.878	0.887	18
mAP (All Classes)	—	—	0.847	216

D. Ablation Study

We conduct a systematic ablation study to quantify the contribution of each proposed component. Results are reported on the validation split (218 images) using mAP@50 as the primary metric.

TABLE V
Ablation Study: Component Contribution to mAP@50 (Validation Set)

Configuration	MHSA	TGU	Synthetic Aug.	mAP@50	Δ mAP
YOLOv8n Baseline	×	×	×	0.735	—
+ Synthetic Aug.	×	×	✓	0.762	+2.7
+ MHSA Only	✓	×	×	0.779	+4.4
+ TGU Only	×	✓	×	0.751	+1.6
+ MHSA + TGU	✓	✓	×	0.811	+7.6
SCRSS-STFF (Full)	✓	✓	✓	0.847	+11.2

The ablation study demonstrates that the MHSA module makes the biggest single-component addition of (+4.4 mAP), to low-contrast cryospheric textures. The TGU offers a complementary +1.6 mAP alone, and a constructive +7.6 mAP when used in combination with MHSA, which means that the benefit of spatial attention is increased with a temporal context. Synthetic augmentation cautiously offers an additional +2.7 mAP when using the full model, as comprehended in the baseline.

E. Inference Speed Analysis

Inference latency is a critical requirement when it comes to real-world deployment. Table VI provides measurements of the speed of inference between hardware configurations tested using the Streamlit application interface.

TABLE VI
Inference Latency per Image (640×640) Across Hardware

Hardware	Preprocessing (ms)	Inference (ms)	Total (ms)
Intel Core i7-1165G7 (CPU)	8.2	25.6	33.8
NVIDIA RTX 3080 (GPU)	2.1	4.3	6.4
Streamlit Cloud (CPU)	9.4	31.2	40.6
Google Colab T4 (GPU)	2.8	5.9	8.7

VI. SYSTEM DEPLOYMENT ARCHITECTURE

A. Streamlit Web Application

The SCRSS is deployed as a production Streamlit web application at <https://saskatoon-cryospheric-road-ml-cj68z2ta6kwlseyyyaytr.streamlit.app/> hosted on Streamlit Community Cloud with Debian Trixie Linux base. The application source code is open-sourced at <https://github.com/Aghawafaabbass/Saskatoon-Cryospheric-Road-ML> under the MIT License.

The application workflow proceeds as follows: (1) The user uploads a JPEG or PNG road image via the file upload widget. (2) The uploaded image is decoded by PIL and converted to a NumPy array. (3) YOLOv8 inference is invoked with conf=0.25 threshold using the best.pt model weights loaded via @st.cache_resource for session persistence. (4) Detection results are rendered as annotated bounding box overlays using OpenCV. (5) Detected class labels are mapped against the hazard keyword set {snow, ice, slush, black ice} to determine overall road safety status. (6) A downloadable safety report is generated as a text file. (7) An interactive Folium map is rendered with a color-coded incident marker at a configurable Saskatoon coordinate pair.

B. Software Dependencies

The application requires the following Python dependencies as specified in requirements.txt: opencv-python-headless (headless OpenCV for server deployment), ultralytics (YOLOv8 inference engine), streamlit (web application framework), pillow (image I/O), numpy (array operations), pandas (data manipulation), folium (geospatial visualization), and streamlit-folium (Folium-Streamlit bridge). Python 3.11 is specified in .python-version for compatibility with Streamlit Cloud’s runtime environment.

C. Model Weights and Versioning

The trained model weights file `best.pt` (YOLOv8 PyTorch serialization format) is stored directly in the repository root at <https://github.com/Aghawafaabbass/Saskatoon-Cryospheric-Road-ML/blob/main/best.pt>. The file is loaded at application startup via `YOLO('best.pt')` and cached for the session lifetime. Model versioning follows semantic versioning (v1.0.0 for the initial release), with future retraining cycles to be managed through Git tags and Streamlit's deployment webhook.

D. Geospatial Incident Mapping

The Folium map component renders an interactive Leaflet.js map centered on Saskatoon's urban core (52.1332°N, 106.6700°W) at zoom level 12. Detection results are mapped as color-coded markers: red markers indicate hazardous road conditions (BI, CS, SL, WF detected), while green markers indicate clear asphalt conditions. This geospatial layer enables municipal traffic authorities to aggregate multiple field reports into a spatial heat map for prioritized maintenance dispatch. Future versions will integrate with the City of Saskatoon Open Data portal [38] for automated weather station correlation.

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

A. Discussion

The experimental findings confirm the hypothesis of meaningful gains by the fusion of spatio-temporal features in the classification of the cryospheric road surfaces. The self-attention mechanism of multi-head self-attention mechanism captures the multi-spatial context necessary to make a discrimination between visually similar surface conditions, especially Black ice against wet asphalt, where local texture features are not good enough discrimination mechanisms. By offering a cross-frame coherence of the temporal gating unit, the differences in detection variability due to motion blur and illumination transient effects on dashcam imaging will be reduced (Nevshuk, 2010, p. 145).

The 12.3% improvement in black ice detection accuracy over baseline YOLOv8n (BI AP@50: 0.721 baseline vs. 0.809 proposed) is particularly noteworthy given that black ice is the most safety-critical cryospheric pathology, responsible for approximately 17% of all Canadian winter road fatalities [39]. The synthetic augmentation pipeline contributes independently to this improvement by providing training signal for the specular reflection patterns characteristic of glazed ice.

B. Limitations

Several limitations of the current system warrant acknowledgment:

- **Single-frame deployment mode:** While the STFF module is designed for temporal sequence processing, the current Streamlit application accepts single uploaded images, limiting the temporal gating benefit to zero-initialized context. Future deployment in video streaming mode will fully activate temporal coherence.
- **Dataset geographic specificity:** The DAWN dataset images are predominantly from Middle Eastern and European environments (sandstorm images from Saudi Arabia, fog from the UK). Sub-arctic Canadian road surface textures, including the distinctive asphalt aggregate used in Saskatchewan, may not be fully represented.
- **Class imbalance:** Clear Asphalt (CA) class measures have only 18 test samples, and Wet Frost (WF) has 29 test samples, which limits statistical reliability of per-class.
- **Absence of friction ground truth:** The system writes labels of categorical road surface classes instead of quantitative estimates of the friction coefficient. Okimonetary Integration should be connected to road friction sensors (e.g., Vaisala DSC111) to be able to directly validate μ regression.

C. Future Work

Several research directions emerge from this work: (1) Extension to video streaming inference with full temporal sequence processing, enabling real-time municipal fleet dashcam integration. (2) Multi-sensor fusion incorporating LiDAR surface reflectivity and NIR camera channels for improved black ice disambiguation. (3) Federated learning across multiple municipal vehicle fleets to continuously update model weights without centralizing sensitive roadway imagery. (4) Integration with Environment and Climate Change Canada’s meteorological API for temperature-aware prior adjustment of class probabilities. (5) Expansion of the training dataset with direct field data collection from Saskatoon City Buses equipped with dashcams, targeting 10,000+ Saskatchewan-specific cryospheric road images.

VIII. CONCLUSION

This paper presented the Saskatoon Cryospheric Road Safety System (SCRSS), a novel framework for autonomous classification of cryospheric road surface pathologies in sub-arctic urban environments. The proposed Spatio-Temporal Feature Fusion (STFF) module, integrating multi-head self-attention and temporal gating within the YOLOv8 detection backbone, achieved mAP@50 of 0.847 on the held-out test set, representing an 11.2 mAP point improvement over the YOLOv8n baseline. The system demonstrated particular gains in black ice detection (+12.3% AP), the most safety-critical winter road hazard category.

The complete system is deployed as a publicly accessible Streamlit web application and open-sourced on GitHub, providing the research community with a deployable baseline for cryospheric road intelligence. The training pipeline, model weights, and configuration files are fully reproducible via the provided Google Colab notebook and GitHub repository links.

Through the increasing pace of connected and autonomous vehicle fleet deployment by sub-arctic cities, there will be a critical infrastructure need in terms of the reliability of perception of cryospheric road conditions. A Computer-effectively, browser-deployed framework, the SCRSS framework fills in the gap between laboratory detection research and operation municipal road safety systems.

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